

Introduction to DevOps

Assignment – 2

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Deliverables:

ACEest_Fitness.py, Jenkins Configuration, Kubernetes YAML files, Sonar Cube reports, PyTest report are included as individual files and in the GitHub Repository

GitHub Repository:

<https://github.com/2024tm93056-ux/ACEestFitness>

Docker Hub Repository:

<https://hub.docker.com/repository/docker/itzzmeakhi/aceestfitness/general>

SonarQube Dashboard:

Includes:

Code Smells: 0

Coverage: 90%

Maintainability: A

Kubernetes Endpoint:

<http://192.168.49.2:30001>

CI/CD Architecture Overview

- GitHub for version control and code collaboration.
- Jenkins for CI/CD automation.
- SonarQube for static code analysis.
- Docker for containerization.
- Docker Hub for image registry.
- Kubernetes (Minikube) for deployment and orchestration.

Challenges Faced & Mitigation

Challenge	Solution
SonarQube authentication errors	Added Jenkins token under Manage Credentials
Docker push permission issues	Used Jenkins credential ID dockerhub-creds
Pipeline failures due to missing tools	Installed via apt install docker.io python3-pip

Automation Outcomes

- Fully automated end-to-end CI/CD pipeline.
- Zero manual intervention in build/deploy stages.
- Integrated code quality gates and automated testing.
- Multiple deployment strategies with rollback support.